

CALVIN SCHROEDER

Denver, CO — 720-503-5044 — calvin.schroeder22@gmail.com

TECHNICAL SUMMARY

Magna Cum Laude Aerospace Engineering graduate with proven experience leading multi-disciplinary, international engineering project teams. Technical specialization in aerospace systems engineering, RF communications, and network architectures, with a strong academic foundation in astrodynamics and trajectory modeling. Proficient full-lifecycle hardware-to-software system integration.

EDUCATION

Bachelor of Science in Aerospace Engineering
Texas A&M University, College Station, TX

May 2026
Cumulative GPA: 3.87

TECHNICAL SKILLS

Engineering Domains: RF Engineering, Network Engineering, Systems Engineering, Astrodynamics, Link Budgets, Real-Time Systems

Programming & Software: Python (NumPy, SciPy, Pandas), C++, MATLAB, ROS, RVIZ, FreeFlyer, Fusion, PlatformIO, LaTeX, Linux (Ubuntu, Pop!_OS, Raspbian)

Hardware & Tools: Microcontrollers (ESP32, Arduino), RF Systems, Git/GitHub, Ollama

ENGINEERING & LEADERSHIP EXPERIENCE

KARURA Martian Rover Team (The Mars Society URC)
U.S. Engineering Lead (May 2025 - June 2026)

May 2023 - June 2026
International Collaboration

- Directed a multi-disciplinary team of 40+ mechanical, electrical, and software engineers executing end-to-end development of an autonomous Martian rover platform for international competition.
- Synchronized technical interfaces and configurations across 6 distinct sub-teams and 2 countries, actively resolving design conflicts and managing complex hardware-to-software baseline allocations.
- Engineered long-range RF communication links utilizing comprehensive link budgets for low-latency remote command operations.

Space Pirates Satellite Servicing Team
Communications and C&DH Sub-Team Lead

August 2025 - May 2026

- Engineered Command & Data Handling (C&DH) and communication subsystems utilizing commercial off-the-shelf (COTS) components to meet strict power, mass, and volumetric budget allocations.
- Formulated high-fidelity RF link budgets to guarantee reliable downlink margins between the satellite and ground infrastructure.
- Developed high-fidelity CAD assemblies of internal electronics packaging and formulated subsystem technical baselines to ensure robust hardware-in-the-loop validation testing.

Students for the Exploration and Development of Space (SEDS)
President

July 2023 - June 2025

- Oversaw 7 officers within the executive leadership team to align strategic direction, budget distributions, and inter-organizational outreach frameworks across the university.
- Interviewed and vetted over 100 design team candidates to allocate specialized talent across active aerospace engineering project pipelines.
- Cultivated partnerships with Texas A&M space research labs and industry organizations via structured networking events and targeted outreach.